

USA-Style Twin Horn

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Here is a twin horn circuit, which can produce mono as well as 'stereo' sounds. There are three suitable sets of frequencies

for a horn of this type and they are 400Hz/500Hz, 500Hz/600Hz and 600Hz/700Hz. The frequency set can be selected and tuned using available presets. The mono output produces a mix of both the frequencies from the



in Fig. 1.

Dual-frequency generator is built around dual-timer IC1. Both the timers are operated in astable mode but generate different frequencies, say 400Hz and 500Hz. First astable multivibrator is built around resistor R1, preset VR1, resistor R2 and capacitor C1. Similarly,

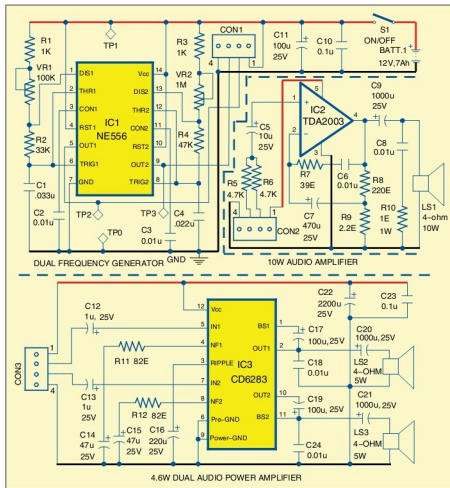


Fig. 1: Circuit diagram of the USA-style twin horn

Test Points

Test point	Details
TP0	0V, GND
TP1	12V
TP2	Frequency set using preset VR1
TP3	Frequency set using preset VR2

PARTS LIST

Semiconductors:

- IC1 - NE555 dual timer
- IC2 - TDA 2003 dual-audio amplifier
- IC3 - CD6283 dual-audio amplifier

Resistors (all 1/4-watt, $\pm 5\%$ carbon, unless stated otherwise):

- R1, R3 - 1-kilo-ohm
- R2 - 33-kilo-ohm
- R4 - 47-kilo-ohm
- R5, R6 - 4.7-kilo-ohm
- R7 - 39-ohm
- R8 - 220-ohm
- R9 - 2.2-ohm
- R10 - 1-ohm, 1W
- R11, R12 - 82-ohm
- VR1 - 10-kilo-ohm preset
- VR2 - 1-mega-ohm preset

Capacitors:

- C1 - 0.033 μ F ceramic disk
- C2, C3, C6, C8, C18, C24 - 0.01 μ F ceramic disk
- C4 - 0.022 μ F ceramic disk
- C5 - 10 μ F, 25V electrolytic
- C7 - 470 μ F, 25V electrolytic
- C9, C20, C21 - 1000 μ F, 25V electrolytic
- C10, C23 - 0.1 μ F ceramic disk
- C11, C17, C19 - 100 μ F, 25V electrolytic
- C12, C13 - 1 μ F, 25V electrolytic
- C14, C15 - 47 μ F, 25V electrolytic
- C16 - 220 μ F, 25V electrolytic
- C22 - 2200 μ F, 25V electrolytic

Miscellaneous:

- CON1-CON3 - 4-pin male and female connector
- BATT.1 - 12V, 7Ah battery
- LS1 - 4-ohm, 10W speaker
- LS2, LS3 - 4-ohm, 5W speaker
- S1 - On/off switch
- Heat sink for TDA2003
- 14-pin IC base

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selected frequency set. The stereo outputs have both frequencies individually on the two channels.

Circuit and description

The circuit is built around dual-timer IC NE555 (IC1), 10W audio amplifier IC TDA2003 (IC2) and 4.6W dual-audio power amplifier CD6283 (IC3).

It has three parts: dual-frequency generator, 10W audio amplifier and 4.6W dual-audio amplifier as shown